



EXAMINATIONS COUNCIL OF ESWATINI
Eswatini General Certificate of Secondary Education

CANDIDATE
NAME

--

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

DESIGN AND TECHNOLOGY

Paper 3 Resistant Materials

6902/03

October/November 2025

1 hour

Candidates answer on the Printed Question Paper.

Additional Materials required: Calculator.

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name in the spaces provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in this section.

Section B

Answer **one** question in this section.

A calculator should be used.

The number of marks is given in brackets [] at the end of each question or part question.

The total marks for this paper is 50.

For Examiner's Use	
Section A	
Section B	
Total	

This document consists of **16** printed pages.

Section A

Answer **all** questions in this section.

- 1 Fig. 1 shows two different cutting tools.

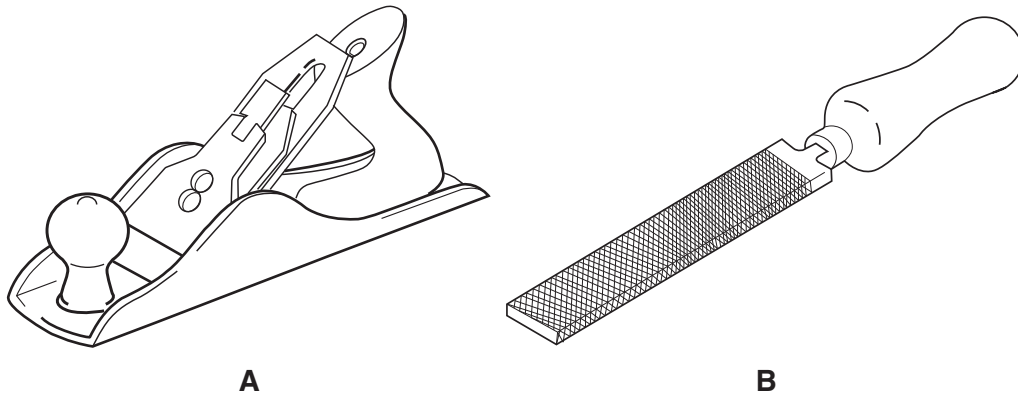


Fig. 1

Name the cutting tools **A** and **B**

A [1]

B [1]

- 2 Complete the table below by naming the tools and stating a specific use for each.

Tool/item of equipment	Name	Specific use
		
		

[4]

- 3 Fig. 2 shows a wall-mounted bench that is used outdoors

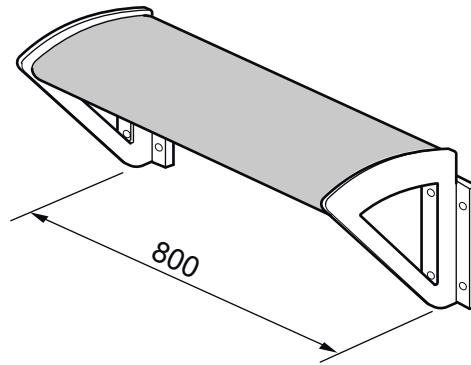


Fig. 2

Explain how anthropometric data is used to decide the height at which the bench could be fixed to a wall.

.....

.....

..... [1]

- 4 Fig. 3 shows a food container made from expanded polystyrene.

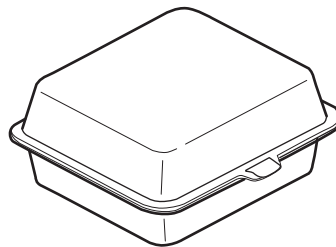


Fig. 3

Give **two** reasons why expanded polystyrene has been used to make the food container.

Reason 1 [1]

Reason 2 [1]

- 5 Fig. 4 shows a saw.

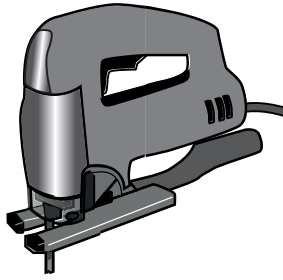


Fig. 4

- (a) Name the saw shown in Fig. 4.

..... [1]

- (b) Give **one** advantage of using the saw in Fig. 4.

..... [1]

- 6 Complete the table below to show a property and use for each of the three materials.

Material	Property	Use
Mahogany		
High carbon steel		
Leather		

[6]

- 7 Fig. 5 shows a tool that is used in a workshop.

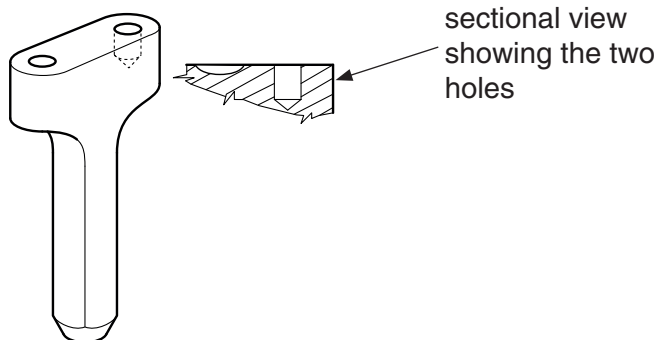


Fig. 5

Explain why the tool shown in Fig. 5 has two different shaped holes.

.....
 [2]

- 8 Annealing is carried out when metal becomes work hardened due to hammering.

Describe how brass sheet would be annealed.

.....

.....

..... [2]

- 9 Fig. 6 shows a knock-down fitting used to assemble wooden parts.

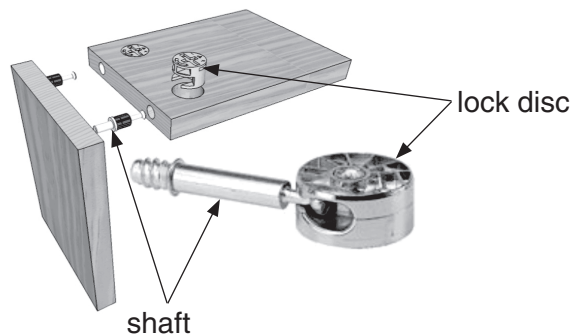


Fig. 6

- (a) Give the correct name for the knock-down fitting shown in Fig. 6.

..... [1]

- (b) Name a piece of equipment that can be used to ensure the holes for the lock disc and shaft align

..... [1]

- 10 Complete the table below to show a specific use for each adhesive.

Adhesive	Specific use
PVA	
Tensol Cement	

[2]

Section B

Answer **one** question in this section.

- 11 Fig. 7 shows a design for a go-kart that is sold as a flat-pack product for self-assembly.

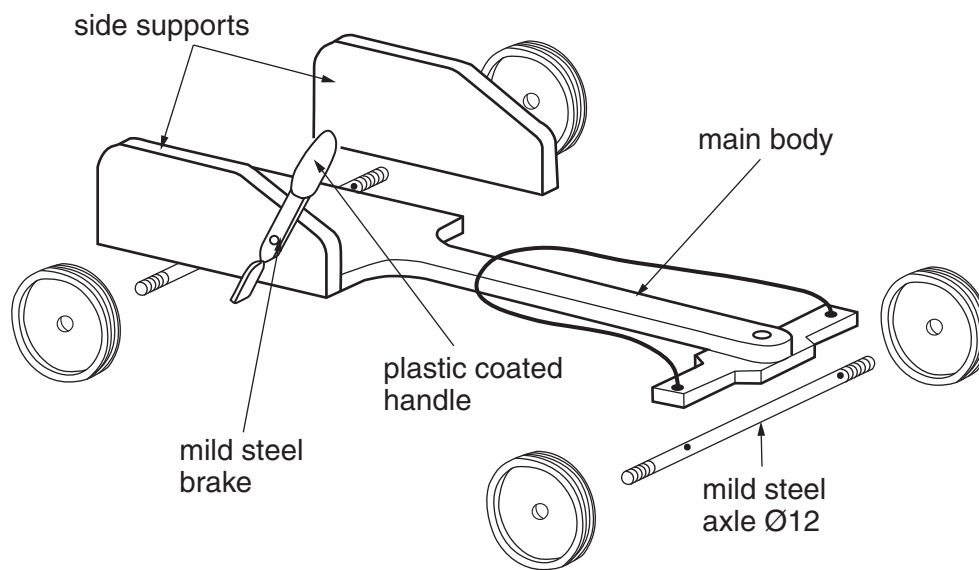


Fig. 7

- (a) Name **two** suitable hardwoods that could be used for the main body of the go-kart.

1 [1]

2 [1]

- (b) State **two** advantages of using solid wood rather than manufactured board for the main body and side supports for the go-kart.

1 [1]

2 [1]

- (c) Give **two** reasons why flat-pack products are very popular with consumers.

1 [1]

2 [1]

- (d) Fig. 8 shows one side support for the go-kart.

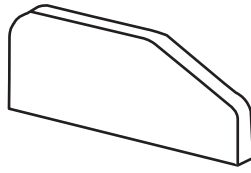


Fig. 8

- (i) Name a portable power saw that could be used to cut out the shape of the side support from a rectangular piece of wood.

..... [1]

- (ii) State **one** safety precaution you would take when using the saw named in (i).

..... [1]

- (iii) Use sketches and notes to show how the side support could be attached to the main body.

[3]

- (e)** The go-kart has a diameter 12 mm mild steel axle which is threaded for 20 mm on both ends.
- (i)** Use sketches and notes to show how to cut the screw thread by hand on one end of the axle. Include preparing the rod end, holding devices, and tools used.

[5]

- (ii)** Use sketches and notes to show how the mild steel brake handle could be coated with plastic to provide a safe and smooth finish.

[4]

- (f) Fig. 9 shows the main body and the side supports that make up the seat of the go-kart.

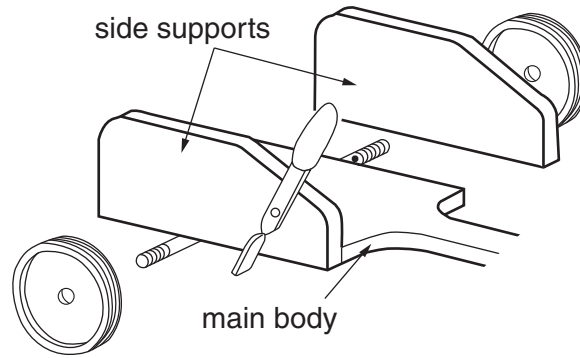


Fig. 9

Use sketches and notes to show modifications to the seat to improve its strength so that the sides cannot collapse easily.

Include details of materials, constructions and fittings used.

12 Fig. 10 shows details of a design for a tooth brush holder.

The holder is made from a thermoplastic sheet and the base from hardwood.

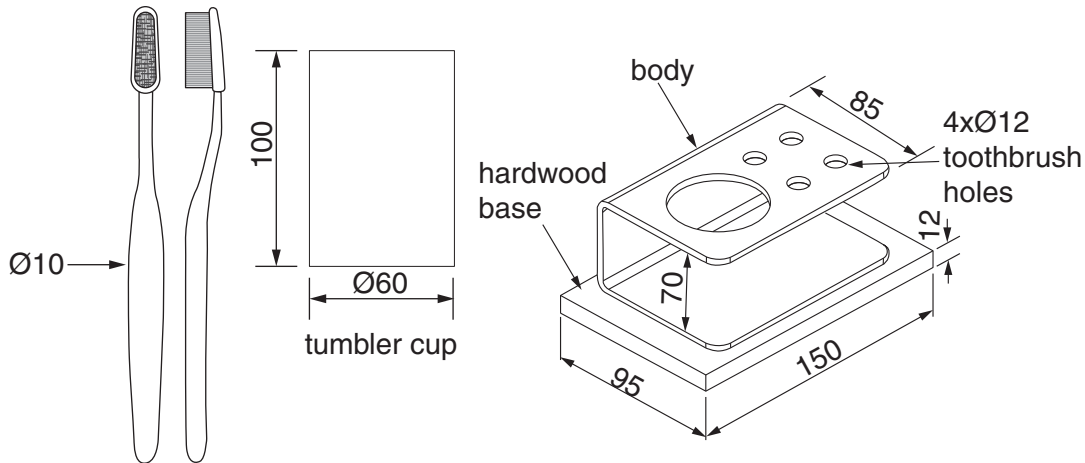


Fig. 10

(a) Give **one** advantage of using a thermoplastic sheet for the toothbrush holder, other than it can be heated and bent to shape.

..... [1]

(b) Name **two** methods of heating the thermoplastic sheet so that it can be bent to shape.

1 [1]

2 [1]

(c) Fig. 11 shows an injection moulding machine that could be used to make the toothbrush holder.

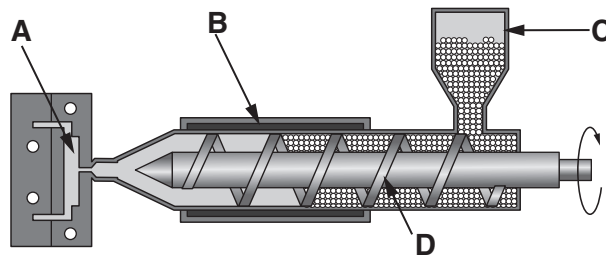


Fig. 11

Name the parts **A**, **B**, **C** and **D** of the machine shown in Fig. 11.

A [1]

B [1]

C [1]

D [1]

- (d)** The body of the toothbrush holder is to be attached to the base and the tumbler cup raised so that it can easily be accessed.

- (i)** Use sketches and notes to show how the body could be fixed to the hardwood base.

[3]

- (ii)** Use sketches and notes to show how the tumbler cup could be raised to enable easy lifting.

[3]

- (iii)** Use sketches and notes to show how to plane the end grain of the base so that the hardwood does not split.

[3]

- (e) The surface of the thermoplastic sheet used for the body of the toothbrush holder is considered to be self-finished.

- (i) State what is meant by the term self-finished.

.....
..... [1]

- (ii) Complete the stages below to show how to finish the edges of the thermoplastic sheet used for the body of the toothbrush holder.

Stage 1 file/plane/sand.

Stage 2 [1]

Stage 3 [1]

- (f) Fig. 12 shows an alternative design for a toothbrush holder made from 3 mm thick mild steel sheet.

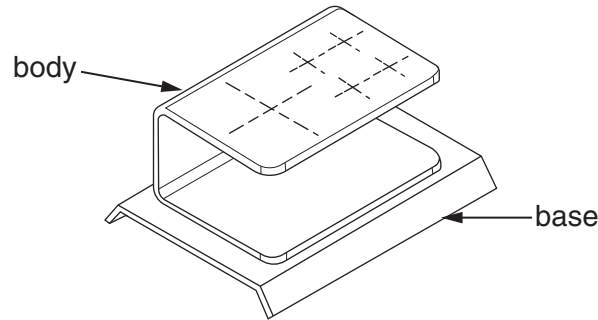


Fig. 12

Use sketches and notes to show how the body could be riveted to the base to produce a flush surface. Name all the tools and equipment used.

[6]

13 Fig. 13 shows a plant holder and the brackets that will support it on a wall.

The sides and ends of the plant holder are made from 3 mm thick mild steel and the base from 10 mm thick acrylic. The brackets are made from mild steel.

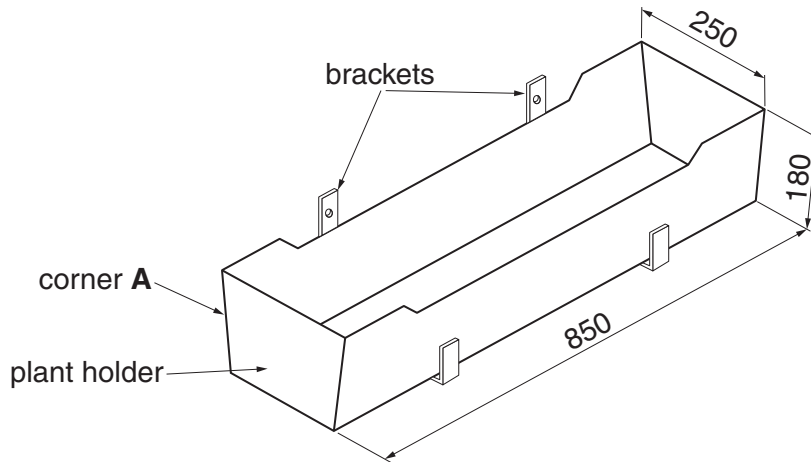


Fig. 13

(a) Give **two** properties of mild steel that makes it suitable for the plant holder.

1 [1]

2 [1]

(b) Use sketches and notes to show how the joint at corner **A** could be welded.

[5]

- (c) Fig. 14 shows the base of the plant holder with Ø3 holes drilled to allow for drainage.

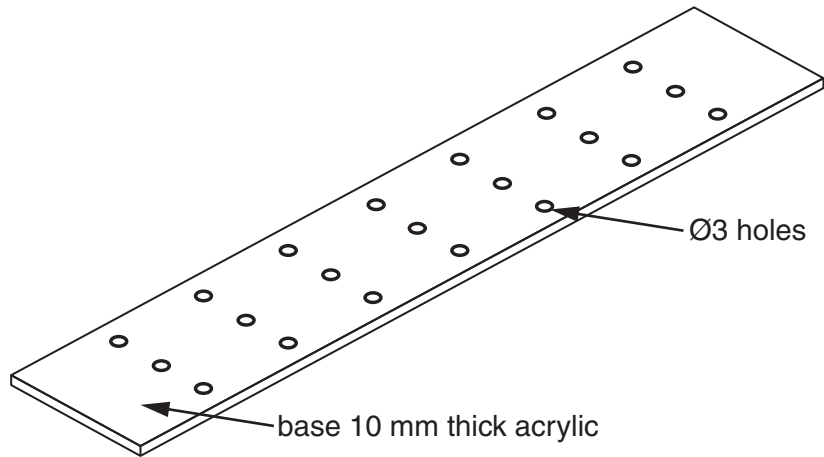


Fig. 14

- (i) Name a specific type of drill bit that could be used to drill the Ø3 holes.

..... [1]

- (ii) Give **two** reasons why the acrylic base would need to be clamped to a piece of scrap wood when drilling the diameter Ø3 holes.

1 [1]

2 [1]

- (d) Fig. 15 shows one bracket and the strip of 3 mm thick mild steel from which the bracket has been made. Holes will be drilled at centres **A** and the strip will be bent at **B**.

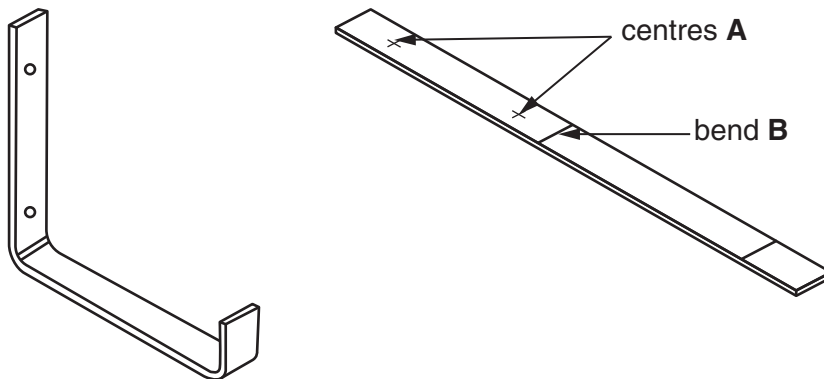


Fig. 15

- (i) Name **two** marking out tools that could be used to mark out centres **A**.

1 [1]

2 [1]

- (ii) Use sketches and notes to show how the mild steel strip could be bent to shape at **B**.

[3]

- (iii) Give **two** reasons why the mild steel brackets would need to be painted.

1 [1]

2 [1]

- (e) The mild steel used to make the plant holder is considered to be a sustainable material.

Explain what is meant by the term 'sustainable'.

.....

.....

..... [1]

- (f) The plant holder could be freestanding rather than wall-mounted.

Fig. 16 shows the plant holder 500 mm above the ground.

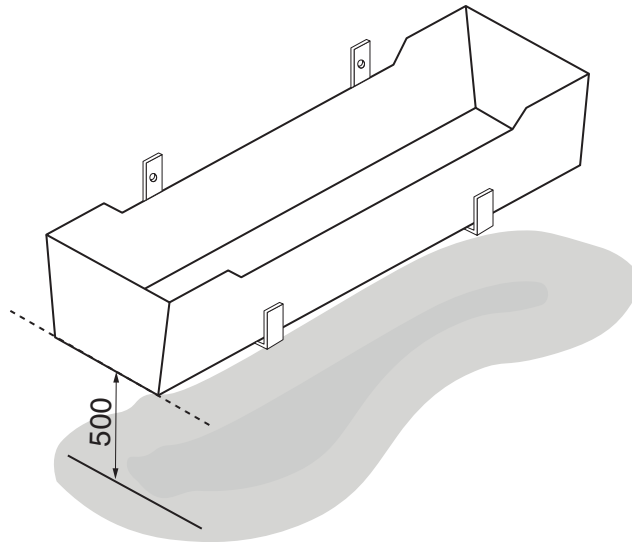


Fig. 16

Use sketches and notes to show how the plant holder could be made freestanding at a height of 500 mm above the ground.

Include details of additional materials, joining methods and sizes.

[7]